Laxmi Charitable Trust’s
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester VI)
Business Analytics
Practical I

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Class : TYIT	Date/Time of Submission : 24/07/2026

Introduction to Business Analytics:
Aim: Collect data from a real-life business scenario and perform exploratory data analysis (EDA) to gain insights into the dataset.
Code:

```
#Import Required Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
#Load dataset
df = pd.read_csv("Retail_sales.csv")
df.head()
```

	Store ID	ID	Date	Sold	(USD)	Percentage	(USD)	Store Location	Category	Week	Effect
0	Spearsland	52372247	2022-01-01	9	2741.69	20	81	Tanzania	Furniture	Saturday	False
1	Spearsland	52372247	2022-01-02	7	2665.53	0	0	Mauritania	Furniture	Sunday	False
2	Spearsland	52372247	2022-01-03	1	380.79	0	0	Saint Pierre and Miquelon	Furniture	Monday	False
3	Spearsland	52372247	2022-01-04	4	1523.16	0	0	Australia	Furniture	Tuesday	False
4	Spearsland	52372247	2022-01-05	2	761.58	0	0	Swaziland	Furniture	Wednesday	False

```
df.tail()
```
✓ 0.0s Python

	Store ID	Product ID	Date	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)	Store Location	Product Category	Day of the Week	Holiday Effect
29995	Spearsland	50239115	2022-01-25	5	2501.15	0	100	Malawi	Clothing	Tuesday	False
29996	Spearsland	50239115	2022-01-26	3	1500.69	0	0	Sudan	Clothing	Wednesday	False
29997	Spearsland	50239115	2022-01-27	6	3001.38	0	0	South Georgia and the South Sandwich Islands	Clothing	Thursday	False
29998	Spearsland	50239115	2022-01-28	5	2501.15	0	0	Haiti	Clothing	Friday	False
29999	Spearsland	50239115	2022-01-29	3	1425.66	5	190	Mozambique	Clothing	Saturday	False

```
df.shape
```
✓ 0.0s

```
(30000, 11)
```

```
#Display Column Names
df.columns
```

✓ 0.0s

```
Index(['Store ID', 'Product ID', 'Date', 'Units Sold', 'Sales Revenue (USD)',
       'Discount Percentage', 'Marketing Spend (USD)', 'Store Location',
       'Product Category', 'Day of the Week', 'Holiday Effect'],
      dtype='str')
```

```
#Display Dataset Information
df.info()
```

✓ 0.0s

```
<class 'pandas.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 11 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Store ID               30000 non-null  str
 1   Product ID             30000 non-null  int64
 2   Date                   30000 non-null  str
 3   Units Sold             30000 non-null  int64
 4   Sales Revenue (USD)    30000 non-null  float64
 5   Discount Percentage    30000 non-null  int64
 6   Marketing Spend (USD)  30000 non-null  int64
 7   Store Location         30000 non-null  str
 8   Product Category       30000 non-null  str
 9   Day of the Week        30000 non-null  str
 10  Holiday Effect         30000 non-null  bool
dtypes: bool(1), float64(1), int64(4), str(5)
memory usage: 2.3 MB
```

```
#Display Data Types
df.dtypes
```

✓ 0.0s

```
Store ID                     str
Product ID                 int64
Date                         str
Units Sold                 int64
Sales Revenue (USD)      float64
Discount Percentage        int64
Marketing Spend (USD)      int64
Store Location               str
Product Category             str
Day of the Week              str
Holiday Effect              bool
dtype: object
```

```
#Check Missing Values
df.isnull().sum()
```
✓ 0.0s

```
Store ID                0
Product ID              0
Date                    0
Units Sold              0
Sales Revenue (USD)     0
Discount Percentage     0
Marketing Spend (USD)   0
Store Location          0
Product Category        0
Day of the Week         0
Holiday Effect          0
dtype: int64
```

#Check Duplicate Records
df.duplicated().sum()

```
#Display Descriptive Statistics
df.describe()
```
✓ 0.0s

	Product ID	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)
count	3.000000e+04	30000.000000	30000.000000	30000.000000	30000.000000
mean	4.461294e+07	6.161967	2749.509593	2.973833	49.944033
std	2.779759e+07	3.323929	2568.639288	5.974530	64.401655
min	3.636541e+06	0.000000	0.000000	0.000000	0.000000
25%	2.228600e+07	4.000000	882.592500	0.000000	0.000000
50%	4.002449e+07	6.000000	1902.420000	0.000000	1.000000
75%	6.559352e+07	8.000000	3863.920000	0.000000	100.000000
max	9.628253e+07	56.000000	27165.880000	20.000000	199.000000

```
#Count Unique Values
df.nunique()
```
✓ 0.0s

```
Store ID                   1
Product ID                42
Date                     731
Units Sold                32
Sales Revenue (USD)     2545
Discount Percentage        5
Marketing Spend (USD)    200
Store Location           243
Product Category           4
Day of the Week            7
Holiday Effect             2
dtype: int64
```

```
#Display Unique Values of a Particular Column
df["Product Category"].unique()
```

✓ 0.0s

```
<StringArray>
['Furniture', 'Electronics', 'Groceries', 'Clothing']
Length: 4, dtype: str
```

```
#Count Product Categories
df["Product Category"].value_counts()
```

✓ 0.0s

```
Product Category
Furniture      9503
Electronics    8041
Clothing       6608
Groceries      5848
Name: count, dtype: int64
```

```
#Count Store Locations
df["Store Location"].value_counts()
```

✓ 0.0s

```
Store Location
Korea                       237
Congo                       234
Turks and Caicos Islands    156
Guam                        153
Tanzania                    151
                           ...
Palestinian Territory        98
Haiti                        96
Dominica                     95
Ireland                      92
Andorra                      80
Name: count, Length: 243, dtype: int64
```

```
#Count Holiday Effect
df["Holiday Effect"].value_counts()
```

✓ 0.0s

```
Holiday Effect
False    29836
True       164
Name: count, dtype: int64
```

```python
#Calculate Average Sales Revenue
df["Sales Revenue (USD)"].mean()
```
✓ 0.0s

```
np.float64(2749.509592666667)
```

```python
#Find Maximum Sales Revenue
df["Sales Revenue (USD)"].max()
```
✓ 0.0s

```
np.float64(27165.88)
```

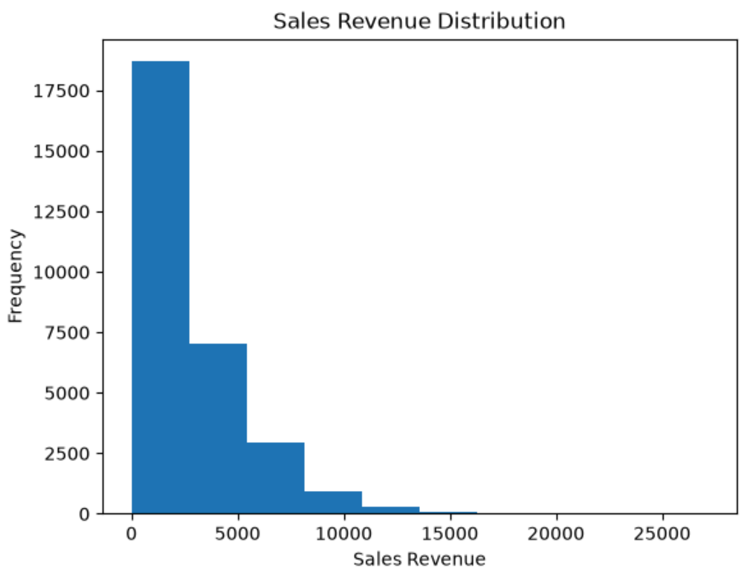
```python
#Find Minimum Sales Revenue
df["Sales Revenue (USD)"].min()
```
✓ 0.0s

```
np.float64(0.0)
```

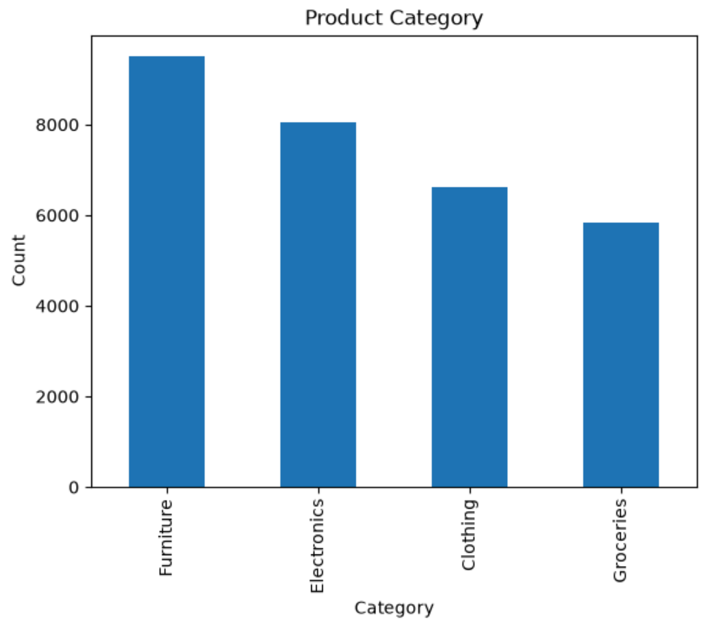
```python
#Correlation Matrix
df.corr(numeric_only=True)
```
✓ 0.0s

	Product ID	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)	Holiday Effect
Product ID	1.000000	-0.008570	-0.181934	-0.006891	-0.002496	-0.000015
Units Sold	-0.008570	1.000000	0.615767	-0.001620	-0.004750	0.159324
Sales Revenue (USD)	-0.181934	0.615767	1.000000	-0.065791	-0.002732	0.090681
Discount Percentage	-0.006891	-0.001620	-0.065791	1.000000	-0.000322	0.005848
Marketing Spend (USD)	-0.002496	-0.004750	-0.002732	-0.000322	1.000000	-0.000455
Holiday Effect	-0.000015	0.159324	0.090681	0.005848	-0.000455	1.000000

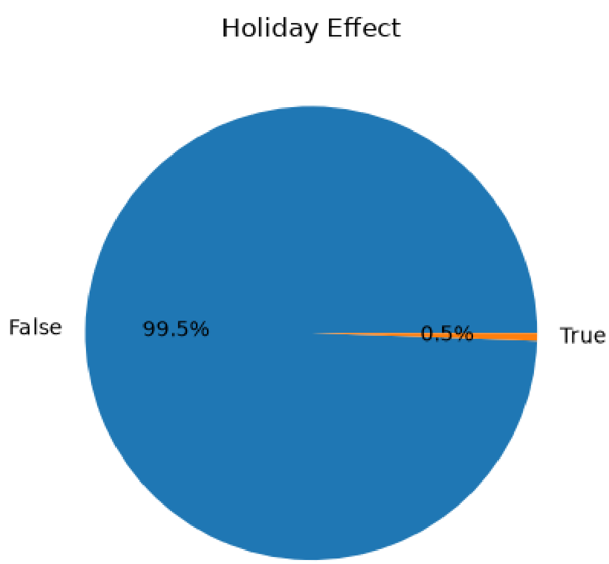
```python
#Histogram
plt.hist(df["Sales Revenue (USD)"])
plt.title("Sales Revenue Distribution")
plt.xlabel("Sales Revenue")
plt.ylabel("Frequency")
plt.show()
```



```
#Bar chart
df["Product Category"].value_counts().plot(kind="bar")
plt.title("Product Category")
plt.xlabel("Category")
plt.ylabel("Count")
plt.show()
```

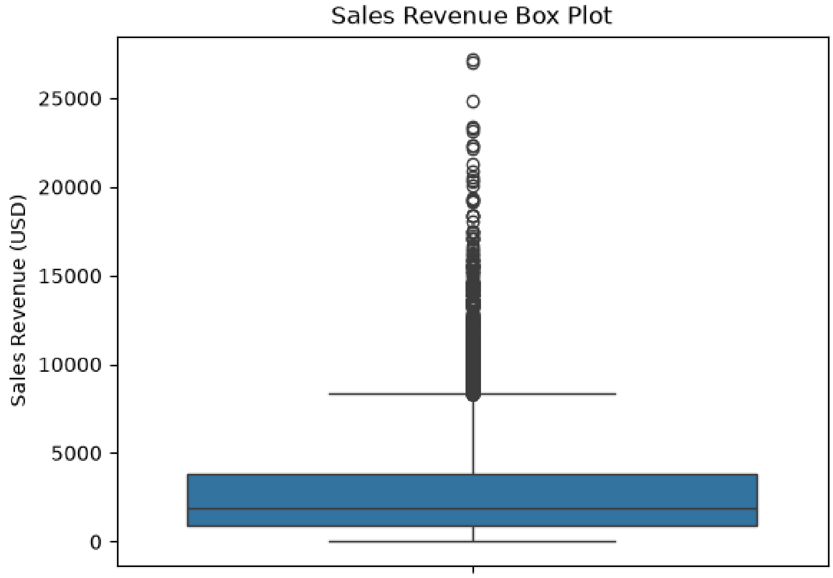



```
#Pie chart
df["Holiday Effect"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.title("Holiday Effect")
plt.ylabel("")
plt.show()
```

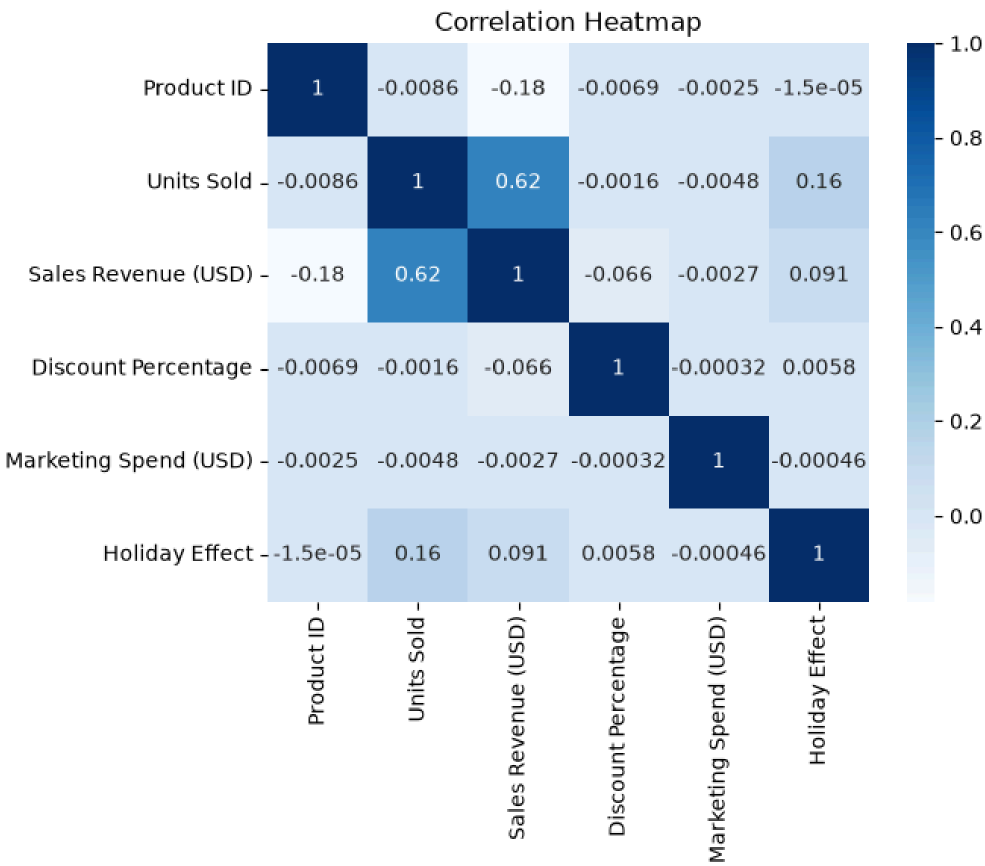



```
#Box plot
sns.boxplot(y=df["Sales Revenue (USD)"])
```

```
plt.title("Sales Revenue Box Plot")
plt.show()
```




```
#Correlation Heatmap
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="Blues")
plt.title("Correlation Heatmap")
plt.show()
```



1(B)

Aim: Analyze customer data to identify trends and patterns that can be used for business decision-making.

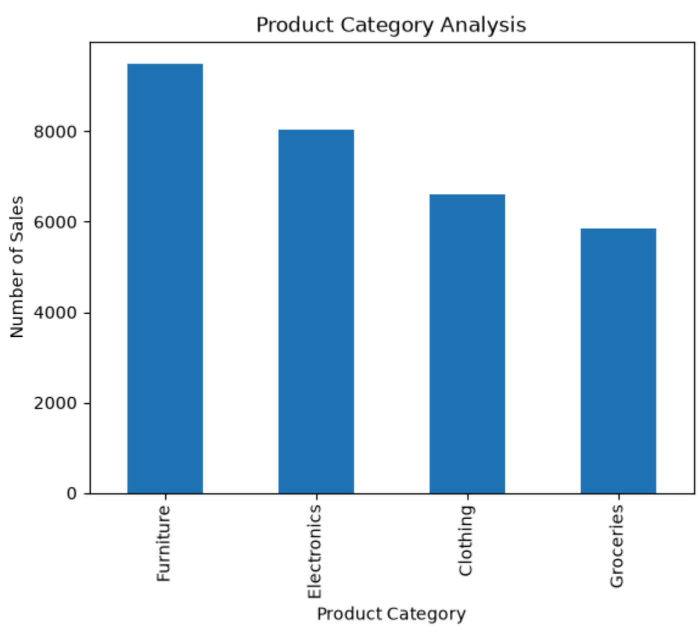
Code:

#Import Required Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

```
#Load dataset
df = pd.read_csv("Retail_sales.csv")
print(df["Product Category"].value_counts())
```

```
Product Category
Furniture      9503
Electronics    8041
Clothing       6608
Groceries      5848
Name: count, dtype: int64
```

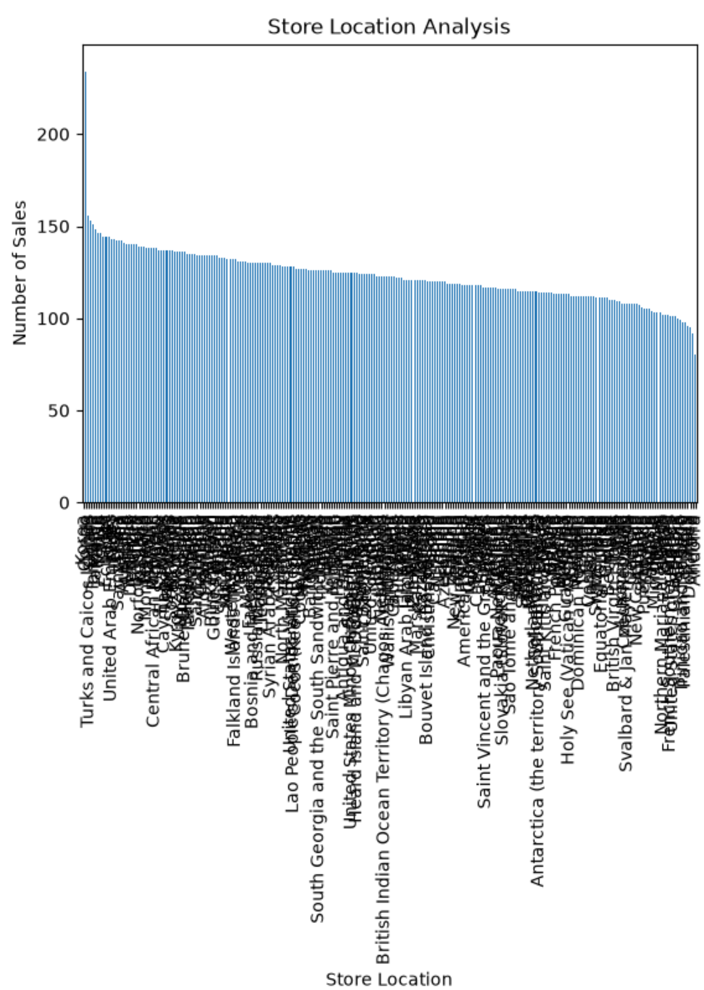
```
df["Product Category"].value_counts().plot(kind="bar")
plt.title("Product Category Analysis")
plt.xlabel("Product Category")
plt.ylabel("Number of Sales")
plt.show()
```




```
print(df["Store Location"].value_counts())
```

```
Store Location
Korea                          237
Congo                          234
Turks and Caicos Islands       156
Guam                           153
Tanzania                       151
                              ...
Palestinian Territory           98
Haiti                           96
Dominica                        95
Ireland                         92
Andorra                         80
Name: count, Length: 243, dtype: int64
```

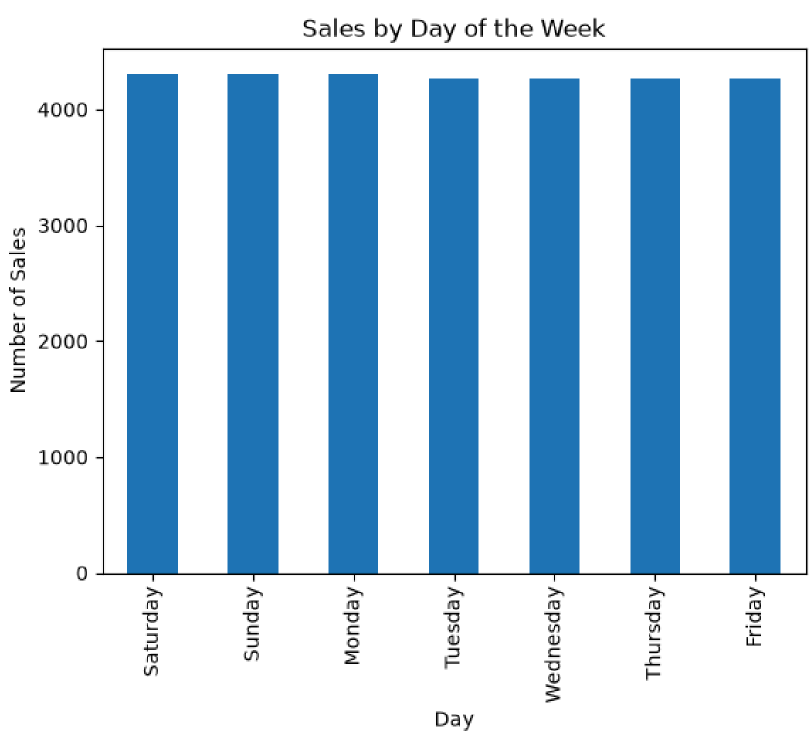
```
df["Store Location"].value_counts().plot(kind="bar")
plt.title("Store Location Analysis")
plt.xlabel("Store Location")
plt.ylabel("Number of Sales")
plt.show()
```



```
print(df["Day of the Week"].value_counts())
```

```
Day of the Week
Saturday     4310
Sunday       4309
Monday       4309
Tuesday      4268
Wednesday    4268
Thursday     4268
Friday       4268
Name: count, dtype: int64
```

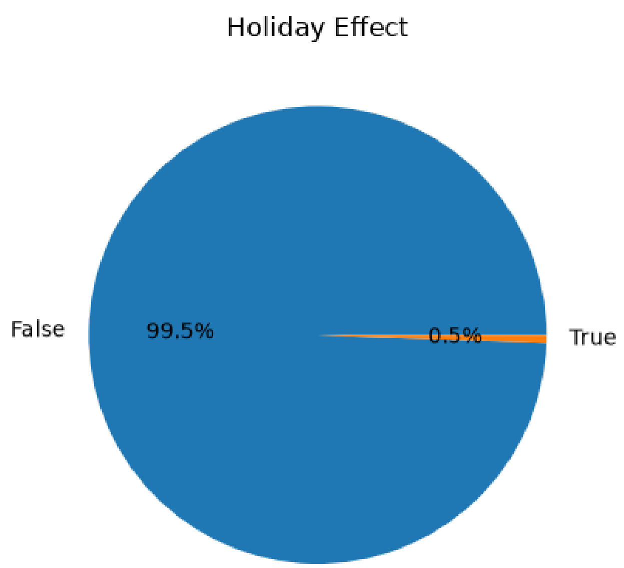
```
df["Day of the Week"].value_counts().plot(kind="bar")
plt.title("Sales by Day of the Week")
plt.xlabel("Day")
plt.ylabel("Number of Sales")
plt.show()
```



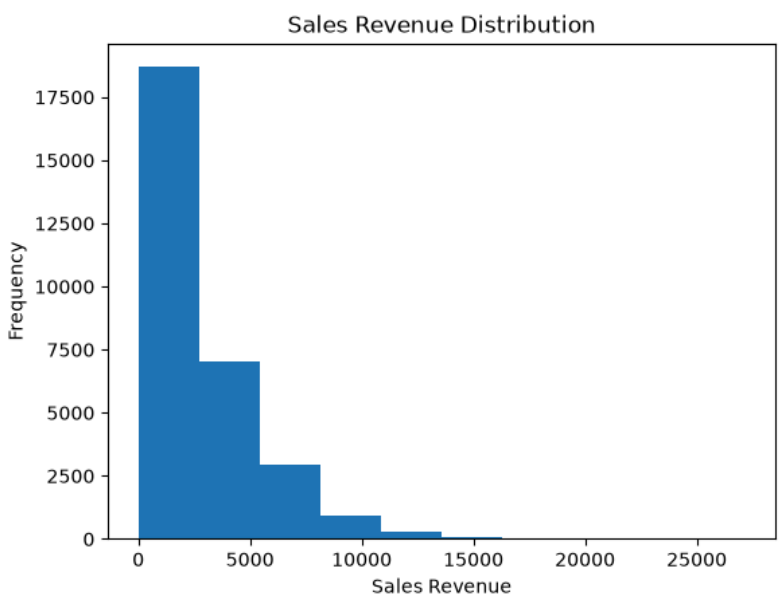

```
print(df["Holiday Effect"].value_counts())
```

```
Holiday Effect
False    29836
True       164
Name: count, dtype: int64
```

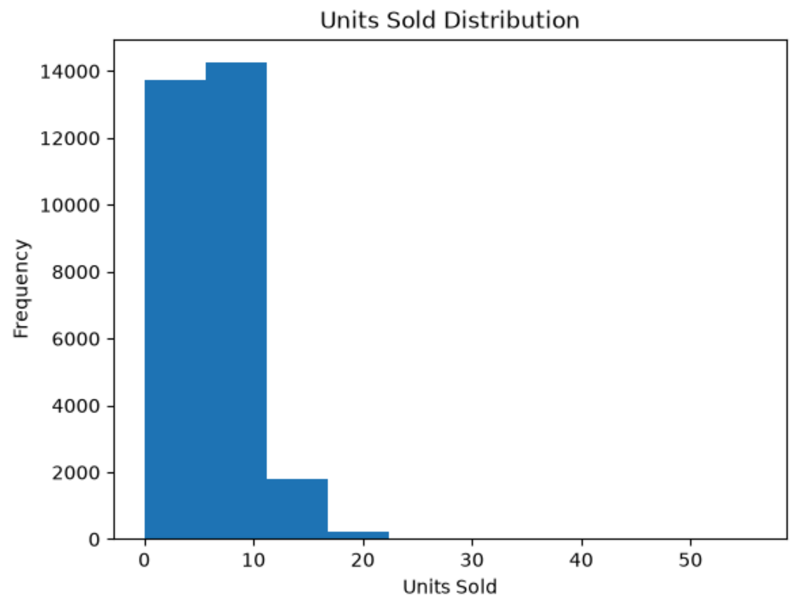
```
df["Holiday Effect"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.title("Holiday Effect")
plt.ylabel("")
plt.show()
```



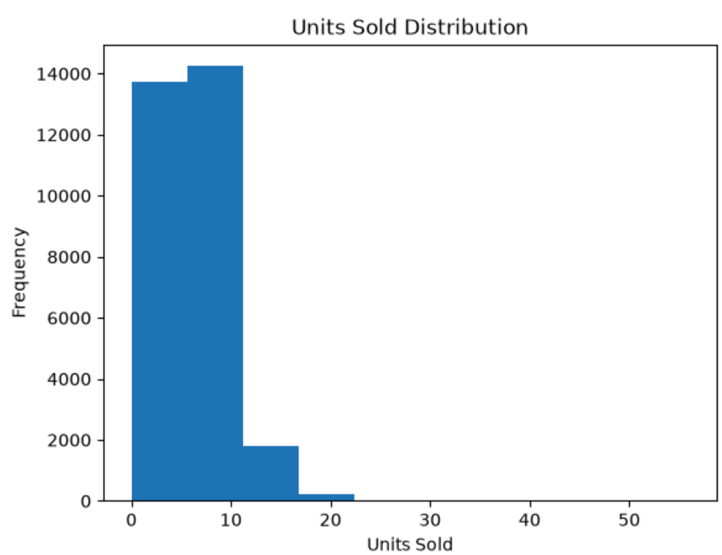

```
plt.hist(df["Sales Revenue (USD)"], bins=10)
plt.title("Sales Revenue Distribution")
plt.xlabel("Sales Revenue")
plt.ylabel("Frequency")
plt.show()
```



```
plt.hist(df["Units Sold"], bins=10)
plt.title("Units Sold Distribution")
plt.xlabel("Units Sold")
plt.ylabel("Frequency")
plt.show()
```




```
plt.hist(df["Units Sold"], bins=10)
plt.title("Units Sold Distribution")
plt.xlabel("Units Sold")
plt.ylabel("Frequency")
plt.show()
```



```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="Oranges")
plt.title("Correlation Heatmap")
plt.show()
```